Indefinite Integration: Formula Sheet

Basic

Power:

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C,\ n \neq -1$$

Reciprocal:

$$\int \frac{dx}{ax+b} = \frac{1}{a}\ln|ax+b|$$

Exponential: $\int e^x dx = e^x$, $\int a^x dx = \dfrac{a^x}{\ln a}$

Sine, cosine: $\int \sin x = -\cos x$, $\int \cos x = \sin x$

Squares: $\int \sec^2 x = \tan x$, $\int \operatorname{cosec}^2 x = -\cot x$

Products: $\int \sec x \tan x = \sec x$, $\int \operatorname{cosec} x \cot x = -\operatorname{cosec} x$

Tan, cot: $\int \tan x = \ln|\sec x|$, $\int \cot x = \ln|\sin x|$

Sec:

$$\int \sec x = \ln|\sec x + \tan x| = \ln\left|\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)\right|$$

Cosec:

$$\int \operatorname{cosec} x = \ln|\operatorname{cosec} x - \cot x| = \ln\left|\tan\frac{x}{2}\right|$$

Algebraic standard forms

$$\int \frac{dx}{a^2+x^2} = \frac{1}{a}\tan^{-1}\frac{x}{a}$$

$$\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right|$$

$$\int \frac{dx}{a^2-x^2} = \frac{1}{2a}\ln\left|\frac{a+x}{a-x}\right|$$

$$\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a}$$

$$\int \frac{dx}{\sqrt{x^2\pm a^2}} = \ln\left|x+\sqrt{x^2\pm a^2}\right|$$

$$\int \frac{dx}{x\sqrt{x^2-a^2}} = \frac{1}{a}\sec^{-1}\frac{|x|}{a}$$

$$\int \sqrt{a^2-x^2}\,dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a}$$

$$\int \sqrt{x^2\pm a^2}\,dx = \frac{x}{2}\sqrt{x^2\pm a^2} \pm \frac{a^2}{2}\ln\left|x+\sqrt{x^2\pm a^2}\right|$$

Substitution and parts

Substitution:

$$\int f(\phi(x))\phi'(x)dx = \int f(t)dt$$

Log form:

$$\int \frac{f'(x)}{f(x)}dx = \ln|f(x)|$$

Power form:

$$\int f^n f'\,dx = \frac{f^{n+1}}{n+1}$$

Parts: $\int uv\,dx = u\int v - \int(u'\int v)dx$ (ILATE)

$$\int e^{ax}\sin bx\,dx = \frac{e^{ax}}{a^2+b^2}(a\sin bx - b\cos bx)$$

$$\int e^{ax}\cos bx\,dx = \frac{e^{ax}}{a^2+b^2}(a\cos bx + b\sin bx)$$

Classic 1: $\int e^x(f+f')dx = e^x f$

Classic 2: $\int(f+xf')dx = xf$

General: $\int e^g(fg'+f')dx = e^g f$

Which substitution

a^2+x^2: $x = a\tan\theta$; a^2-x^2: $x = a\sin\theta$; x^2-a^2: $x = a\sec\theta$

$\sqrt{\frac{a-x}{a+x}}$: $x = a\cos 2\theta$; $\sqrt{\frac{x}{a-x}}$: $x = a\sin^2\theta$

$(x-\alpha)(\beta-x)$:

$$x = \alpha\cos^2\theta + \beta\sin^2\theta$$

$(x-\alpha)(x-\beta)$:

$$x = \alpha\sec^2\theta - \beta\tan^2\theta$$

$\dfrac{1}{(ax+b)\sqrt{px+q}}$: $px+q = t^2$; $\dfrac{1}{(ax+b)\sqrt{Q}}$: $ax+b = \frac{1}{t}$

$\dfrac{1}{(ax^2+b)\sqrt{px^2+q}}$: $x = \frac{1}{t}$

$\dfrac{1}{(x-a)^m(x-b)^n}$, $m+n = 2$:

$$\frac{x-a}{x-b} = t$$

Kutur putur: $1+x^{-n} = t$

$\dfrac{x^2\pm k}{x^4+px^2+k^2}$:

$$x \mp \frac{k}{x} = t$$

Trigonometric integrands

Power reduction: $\sin^2 x = \frac{1-\cos 2x}{2}$, $\cos^2 x = \frac{1+\cos 2x}{2}$

Cubes: $\sin^3 x = \frac{3\sin x - \sin 3x}{4}$, $\cos^3 x = \frac{3\cos x + \cos 3x}{4}$

$\sin^m x\cos^n x$: odd power $\to$ other function $= t$; $m+n$ neg. even $\to \tan x = t$

$$\frac{1}{a\sin x + b\cos x} = \frac{1}{r\sin(x+\alpha)},\ r = \sqrt{a^2+b^2}$$

$\dfrac{1}{a\sin^2 x + b\cos^2 x + \cdots}$: $\tan x = t$; $\dfrac{1}{a\sin x + b\cos x + c}$: $\tan\frac{x}{2} = t$

$\mathrm{Nr} = A\,\mathrm{Dr} + B\,\mathrm{Dr}' + C$ for $\dfrac{l\sin x + m\cos x + n}{a\sin x + b\cos x + c}$

$\dfrac{1}{1+\sin x}$: $\tan x - \sec x$; $\dfrac{1}{1+\cos x}$: $\tan\frac{x}{2}$

$$\sin 2x = 1 - (\sin x - \cos x)^2 = (\sin x + \cos x)^2 - 1$$

Partial fractions

Distinct linear:

$$\frac{A}{x-a}+\frac{B}{x-b}$$

Repeated:

$$\frac{A_1}{x-a}+\frac{A_2}{(x-a)^2}+\cdots+\frac{A_k}{(x-a)^k}$$

Quadratic factor:

$$\frac{Bx+C}{x^2+bx+c}$$

Linear over quadratic: linear $= A\,Q' + B$